Name: ___________________________________

9.1 Pythagorean Theorem.

1. Legs 9 and 12: find the hypotenuse. Then legs 6 and 10: find it in simplest radical form.

2. Hypotenuse 22, leg 10: find the other leg in simplest radical form.

3. Legs 8.2 and 5.7: find the hypotenuse to the nearest tenth.

4. Classify: 9, 11, 15. Then classify: 12, 16, 20.

9.2 Special right triangles (exact answers).

5. 45°-45°-90° with legs 14: find the hypotenuse. With hypotenuse 20: find each leg.

6. 30°-60°-90° with shorter leg 6: find the other sides. With longer leg 18: find the other sides.

7. A square has diagonal 18: find its side. An equilateral triangle has side 20: find its altitude.

9.4–9.5 Trig ratios.

8. Right △ with legs 7 and 24, hypotenuse 25; θ is opposite the 7. Write sin θ, cos θ, tan θ.

9. A 47° angle has hypotenuse 26: find the opposite side (nearest tenth).

10. A 34° angle has opposite side 15: find the adjacent side (nearest tenth).

Elevation and depression.

11. From 210 ft away, the angle of elevation to a rooftop is 41°. Find the height (nearest foot).

12. A drone 260 ft up sights a landing pad at a 12° angle of depression. Find the horizontal distance (nearest foot).

13. A watchtower guard 55 ft up sees two hikers in line at depressions of 46° and 21°. How far apart are the hikers (nearest foot)?

9.6 Solving right triangles (decimals to the nearest tenth).

14. Solve: legs 10 and 24.

15. Solve: hypotenuse 34, leg 16.

16. Solve: a 37° angle with hypotenuse 40.

17. Find the angle whose adjacent leg is 20 and hypotenuse is 32 (nearest tenth).

Putting it together.

18. A 16-ft ladder makes a 68° angle with the ground. How high does it reach, and how far is its base from the wall (nearest tenth)?

19. Which tool and why: find the exact hypotenuse of a right triangle with a 60° angle and shorter leg 9.

20. Why can't a right triangle have sides 5, 5, and $5\sqrt{3}$? Check with the converse.